



Kelas : 1H
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Jobsheet Ke- : 13

Praktikum Ke-1

Langkah

Jawaban/Deskripsi

1

The screenshot displays the MySQL Workbench interface. The top pane shows two SQL queries executed against a database named 'shopping06'.

Query 1:

```
ENGINE=InnoDB AUTO_INCREMENT=15 DEFAULT CHARSET=utf8mb4;  
INSERT INTO `nilai` (`id`, `nim`, `kode_matakuliah`, `nilai_huruf`, `nilai_angka`) VALUES  
(1,171401,'ASD','A',85),  
(2,171401,'BD','B',75),  
(3,171401,'PBO','A',90),  
(4,171401,'WEB','B',78),  
(5,171402,'ASD','B',72),  
(6,171402,'BD','C',65),  
(7,171402,'PBO','B',74),  
(8,171403,'ASD','A',92),  
(9,171403,'BD','A',88),  
(10,171403,'PBO','B',76),  
(11,171403,'WEB','A',91),  
(12,171403,'ML','B',80),  
(13,171404,'ASD','C',62),  
(14,171404,'BD','B',73);
```

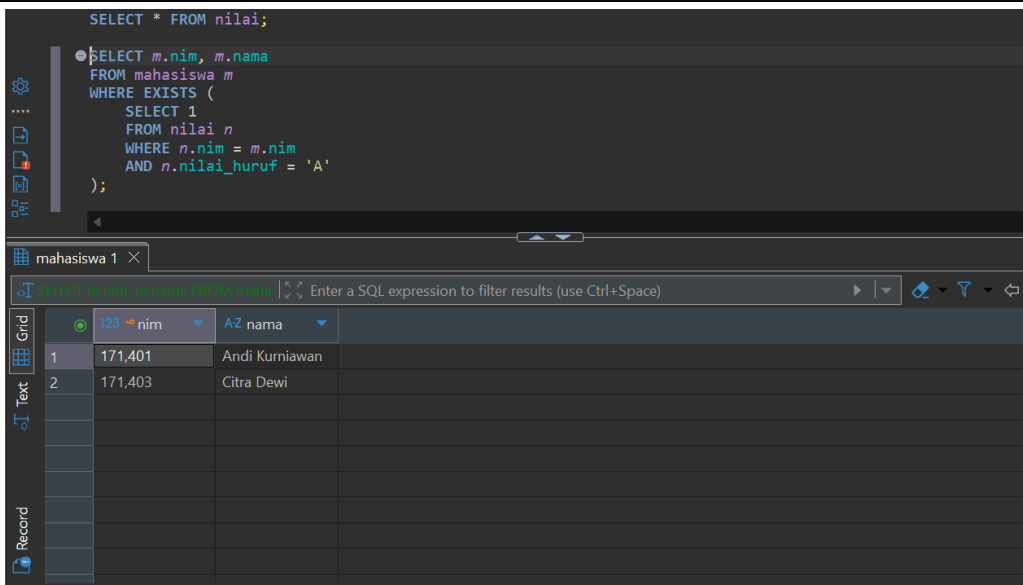
Query 2:

```
SELECT * FROM mahasiswa;  
SELECT * FROM matakuliah;  
SELECT * FROM nilai;
```

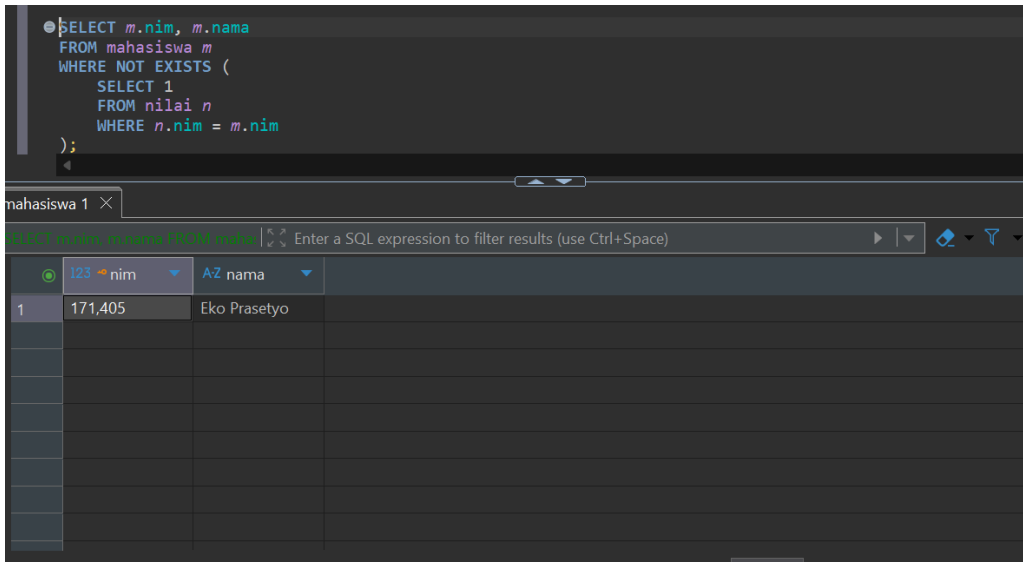
The bottom pane shows the result set for the first query, displaying student information from the 'mahasiswa' table:

#	nim	AZ nama
1	171.401	Andi Kurniawan
2	171.402	Budi Santoso
3	171.403	Citra Dewi
4	171.404	Dian Pratiwi
5	171.405	Eko Prasetyo

Praktikum Ke-2

Langkah	Jawaban/Deskripsi									
1	 The screenshot shows a SQL IDE interface. The top pane displays a SQL query: <pre>SELECT * FROM nilai; SELECT m.nim, m.nama FROM mahasiswa m WHERE EXISTS (SELECT 1 FROM nilai n WHERE n.nim = m.nim AND n.nilai_huruf = 'A');</pre> The bottom pane shows the results of the query in a table view. The table has two columns: 'nim' and 'nama'. The results are as follows: <table border="1"><thead><tr><th></th><th>123 nim</th><th>Az nama</th></tr></thead><tbody><tr><td>1</td><td>171,401</td><td>Andi Kurniawan</td></tr><tr><td>2</td><td>171,403</td><td>Citra Dewi</td></tr></tbody></table>		123 nim	Az nama	1	171,401	Andi Kurniawan	2	171,403	Citra Dewi
	123 nim	Az nama								
1	171,401	Andi Kurniawan								
2	171,403	Citra Dewi								

Praktikum Ke-3

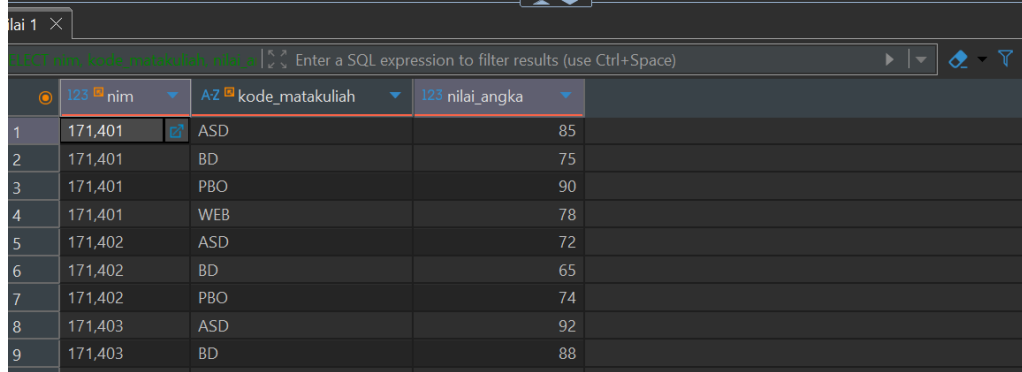
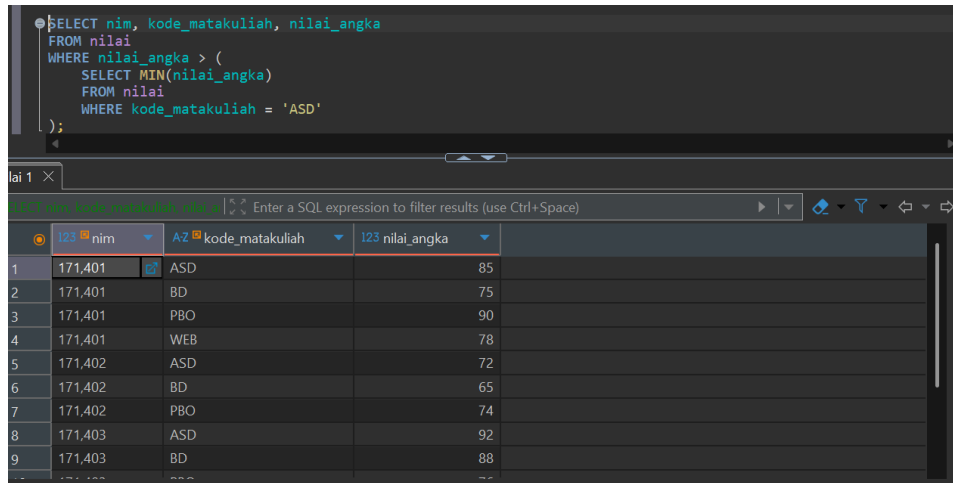
Langkah	Jawaban/Deskripsi						
1	Tampilkan Mahasiswa yang Belum Memiliki Nilai Sama Sekali  <pre>SELECT m.nim, m.nama FROM mahasiswa m WHERE NOT EXISTS (SELECT 1 FROM nilai n WHERE n.nim = m.nim);</pre> mahasiswa 1 <table><thead><tr><th></th><th>123 nim</th><th>AZ nama</th></tr></thead><tbody><tr><td>1</td><td>171,405</td><td>Eko Prasetyo</td></tr></tbody></table>		123 nim	AZ nama	1	171,405	Eko Prasetyo
	123 nim	AZ nama					
1	171,405	Eko Prasetyo					
2	Tampilkan Mata Kuliah yang Belum Ada Mahasiswa yang Mengambilnya						

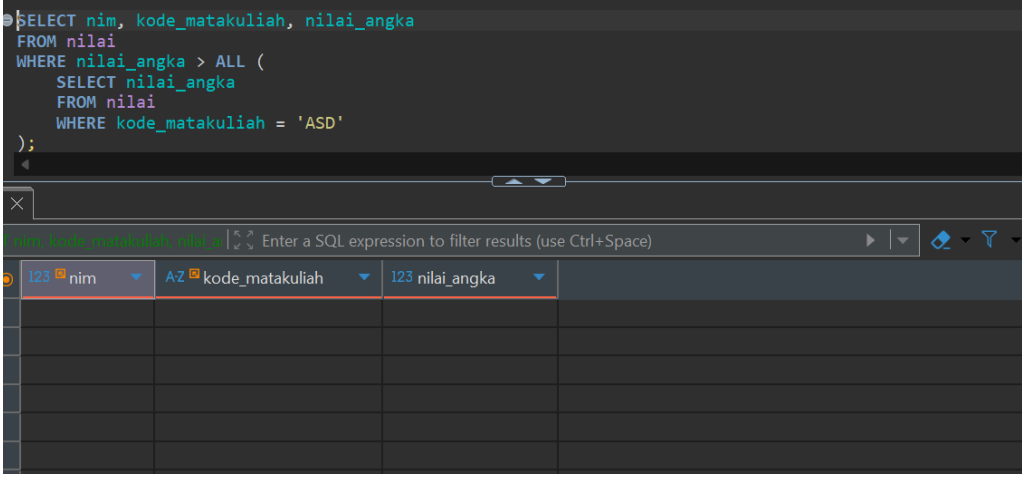
The screenshot shows a SQL IDE interface. The top pane contains a SQL query:

```
SELECT mk.kode, mk.nama
FROM matakuliah mk
WHERE NOT EXISTS (
    SELECT 1
    FROM nilai n
    WHERE n.kode_matakuliah = mk.kode
);
```

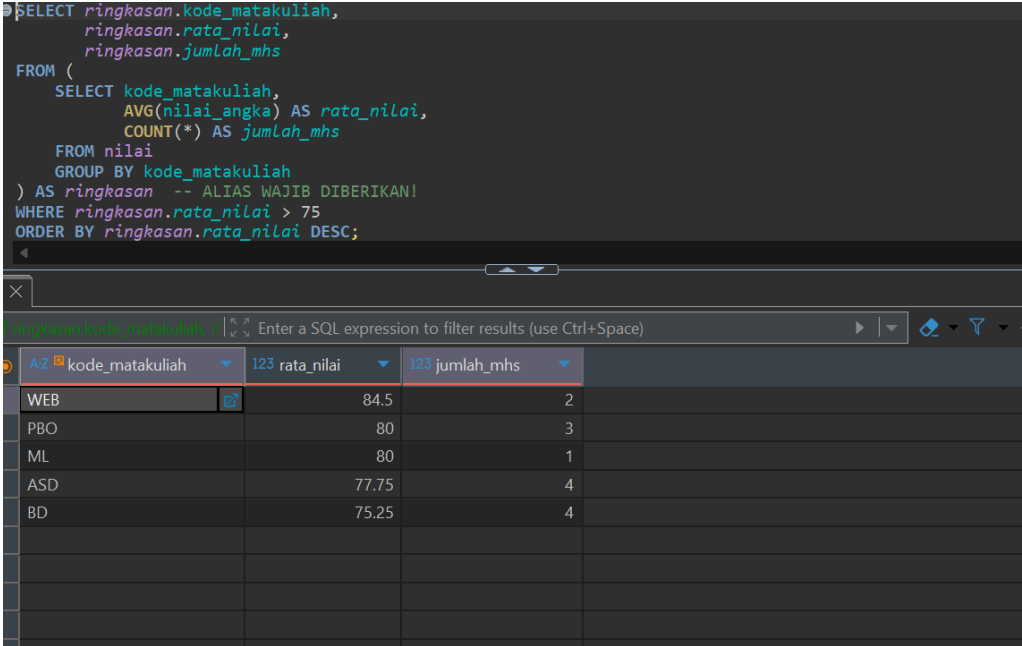
The bottom pane shows the results of the query. The first row is highlighted, showing columns "AZ → kode" and "AZ nama". Below this are several empty rows.

Praktikum Ke-4

Langkah	Jawaban/Deskripsi																																								
1.	Gunakan ANY untuk Perbandingan Nilai <pre>SELECT nim, kode_matakuliah, nilai_angka FROM nilai WHERE nilai_angka > ANY (SELECT nilai_angka FROM nilai WHERE kode_matakuliah = 'ASD');</pre>  <table><tr><th></th><th>123 nim</th><th>AZ kode_matakuliah</th><th>123 nilai_angka</th></tr><tr><td>1</td><td>171,401</td><td>ASD</td><td>85</td></tr><tr><td>2</td><td>171,401</td><td>BD</td><td>75</td></tr><tr><td>3</td><td>171,401</td><td>PBO</td><td>90</td></tr><tr><td>4</td><td>171,401</td><td>WEB</td><td>78</td></tr><tr><td>5</td><td>171,402</td><td>ASD</td><td>72</td></tr><tr><td>6</td><td>171,402</td><td>BD</td><td>65</td></tr><tr><td>7</td><td>171,402</td><td>PBO</td><td>74</td></tr><tr><td>8</td><td>171,403</td><td>ASD</td><td>92</td></tr><tr><td>9</td><td>171,403</td><td>BD</td><td>88</td></tr></table>		123 nim	AZ kode_matakuliah	123 nilai_angka	1	171,401	ASD	85	2	171,401	BD	75	3	171,401	PBO	90	4	171,401	WEB	78	5	171,402	ASD	72	6	171,402	BD	65	7	171,402	PBO	74	8	171,403	ASD	92	9	171,403	BD	88
	123 nim	AZ kode_matakuliah	123 nilai_angka																																						
1	171,401	ASD	85																																						
2	171,401	BD	75																																						
3	171,401	PBO	90																																						
4	171,401	WEB	78																																						
5	171,402	ASD	72																																						
6	171,402	BD	65																																						
7	171,402	PBO	74																																						
8	171,403	ASD	92																																						
9	171,403	BD	88																																						
2.	Ekuivalensi dengan > MIN(...) <pre>SELECT nim, kode_matakuliah, nilai_angka FROM nilai WHERE nilai_angka > (SELECT MIN(nilai_angka) FROM nilai WHERE kode_matakuliah = 'ASD');</pre>  <table><tr><th></th><th>123 nim</th><th>AZ kode_matakuliah</th><th>123 nilai_angka</th></tr><tr><td>1</td><td>171,401</td><td>ASD</td><td>85</td></tr><tr><td>2</td><td>171,401</td><td>BD</td><td>75</td></tr><tr><td>3</td><td>171,401</td><td>PBO</td><td>90</td></tr><tr><td>4</td><td>171,401</td><td>WEB</td><td>78</td></tr><tr><td>5</td><td>171,402</td><td>ASD</td><td>72</td></tr><tr><td>6</td><td>171,402</td><td>BD</td><td>65</td></tr><tr><td>7</td><td>171,402</td><td>PBO</td><td>74</td></tr><tr><td>8</td><td>171,403</td><td>ASD</td><td>92</td></tr><tr><td>9</td><td>171,403</td><td>BD</td><td>88</td></tr></table>		123 nim	AZ kode_matakuliah	123 nilai_angka	1	171,401	ASD	85	2	171,401	BD	75	3	171,401	PBO	90	4	171,401	WEB	78	5	171,402	ASD	72	6	171,402	BD	65	7	171,402	PBO	74	8	171,403	ASD	92	9	171,403	BD	88
	123 nim	AZ kode_matakuliah	123 nilai_angka																																						
1	171,401	ASD	85																																						
2	171,401	BD	75																																						
3	171,401	PBO	90																																						
4	171,401	WEB	78																																						
5	171,402	ASD	72																																						
6	171,402	BD	65																																						
7	171,402	PBO	74																																						
8	171,403	ASD	92																																						
9	171,403	BD	88																																						

3.	Gunakan ALL untuk Perbandingan Nilai <pre> SELECT nim, kode_matakuliah, nilai_angka FROM nilai WHERE nilai_angka > ALL (SELECT nilai_angka FROM nilai WHERE kode_matakuliah = 'ASD'); </pre> 
----	---

Praktikum Ke-5

Langkah	Jawaban/Deskripsi
1.	Filter Hasil Agregasi Menggunakan Derived Table <pre> SELECT ringkasan.kode_matakuliah, ringkasan.rata_nilai, ringkasan.jumlah_mhs FROM (SELECT kode_matakuliah, AVG(nilai_angka) AS rata_nilai, COUNT(*) AS jumlah_mhs FROM nilai GROUP BY kode_matakuliah) AS ringkasan -- ALIAS WAJIB DIBERIKAN! WHERE ringkasan.rata_nilai > 75 ORDER BY ringkasan.rata_nilai DESC; </pre>  ☐ Coba hilangkan alias ringkasan setelah tanda kurung tutup sub-query. Apa yang terjadi? Jawaban: Query akan langsung <i>error</i> (biasanya memunculkan pesan seperti <i>Every derived table must have its own alias</i>). Hal ini terjadi karena SQL membutuhkan pengenalan atau nama tabel sementara tersebut agar kolom-kolom di dalamnya dapat dirujuk oleh outer query. ☐ Mengapa filter <code>rata_nilai > 75</code> tidak bisa langsung ditulis menggunakan HAVING pada query dalam sub-query?

	Jawaban: Sebenarnya operasi tersebut <i>bisa</i> menggunakan HAVING jika langsung di query dalam. Namun, esensi dari praktikum ini adalah menunjukkan bahwa dengan <i>Derived Table</i>, kita bisa memperlakukan hasil agregasi layaknya tabel fisik biasa. Ini memberikan fleksibilitas tinggi jika ke depannya kita ingin melakukan operasi kompleks lain (seperti <i>window function</i> atau pemisahan logika query) yang tidak bisa diselesaikan hanya dengan HAVING biasa.																																								
2.	Gabungkan Derived Table dengan Tabel Lain <pre>SELECT mk.nama AS nama_matakuliah, r.rata_nilai, r.jumlah_mhs FROM (SELECT kode_matakuliah, AVG(nilai_angka) AS rata_nilai, COUNT(*) AS jumlah_mhs FROM nilai GROUP BY kode_matakuliah) AS r JOIN matakuliah mk ON mk.kode = r.kode_matakuliah ORDER BY r.rata_nilai DESC;</pre> matakuliah 1 SELECT mk.nama AS nama_matakuliah; Enter a SQL expression to filter results (use Ctrl+Space) <table><thead><tr><th></th><th>AZ nama_matakuliah</th><th>123 rata_nilai</th><th>123 jumlah_mhs</th></tr></thead><tbody><tr><td>1</td><td>Pemrograman Web</td><td>84.5</td><td>2</td></tr><tr><td>2</td><td>Pemrog. Berorientasi Objek</td><td>80</td><td>3</td></tr><tr><td>3</td><td>Machine Learning</td><td>80</td><td>1</td></tr><tr><td>4</td><td>Algoritma & Struktur Data</td><td>77.75</td><td>4</td></tr><tr><td>5</td><td>Basis Data</td><td>75.25</td><td>4</td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td></tr></tbody></table>		AZ nama_matakuliah	123 rata_nilai	123 jumlah_mhs	1	Pemrograman Web	84.5	2	2	Pemrog. Berorientasi Objek	80	3	3	Machine Learning	80	1	4	Algoritma & Struktur Data	77.75	4	5	Basis Data	75.25	4																
	AZ nama_matakuliah	123 rata_nilai	123 jumlah_mhs																																						
1	Pemrograman Web	84.5	2																																						
2	Pemrog. Berorientasi Objek	80	3																																						
3	Machine Learning	80	1																																						
4	Algoritma & Struktur Data	77.75	4																																						
5	Basis Data	75.25	4																																						

Praktikum Ke-6

Langkah	Jawaban/Deskripsi																								
1.	Tampilkan Nilai Tertinggi Per Mata Kuliah <pre>SELECT n1.nim, n1.kode_matakuliah, n1.nilai_angka FROM nilai n1 WHERE n1.nilai_angka = (SELECT MAX(n2.nilai_angka) FROM nilai n2 WHERE n2.kode_matakuliah = n1.kode_matakuliah -- Titik Korelasi!) ORDER BY n1.kode_matakuliah;</pre> ai 1 X SELECT n1.nim, n1.kode_matakuliah ^ ^ Enter a SQL expression to filter results (use Ctrl+Space) <table><thead><tr><th></th><th>123 nim</th><th>AZ kode_matakuliah</th><th>123 nilai_angka</th></tr></thead><tbody><tr><td>1</td><td>171,403</td><td>ASD</td><td>92</td></tr><tr><td>2</td><td>171,403</td><td>BD</td><td>88</td></tr><tr><td>3</td><td>171,403</td><td>ML</td><td>80</td></tr><tr><td>4</td><td>171,401</td><td>PBO</td><td>90</td></tr><tr><td>5</td><td>171,403</td><td>WEB</td><td>91</td></tr></tbody></table>		123 nim	AZ kode_matakuliah	123 nilai_angka	1	171,403	ASD	92	2	171,403	BD	88	3	171,403	ML	80	4	171,401	PBO	90	5	171,403	WEB	91
	123 nim	AZ kode_matakuliah	123 nilai_angka																						
1	171,403	ASD	92																						
2	171,403	BD	88																						
3	171,403	ML	80																						
4	171,401	PBO	90																						
5	171,403	WEB	91																						

2.

Tampilkan Nilai Maksimum Mata Kuliah Sebagai Kolom Tambahan

```

SELECT n.nim,
       n.kode_matakuliah,
       n.nilai_angka,
       (SELECT MAX(n2.nilai_angka)
        FROM nilai n2
        WHERE n2.kode_matakuliah = n.kode_matakuliah) AS nilai_max_mk
FROM nilai n
ORDER BY n.kode_matakuliah, n.nilai_angka DESC;

```

123 nim	AZ kode_matakuliah	123 nilai_angka	123 nilai_max_mk
171,403	ASD	92	92
171,401	ASD	85	92
171,402	ASD	72	92
171,404	ASD	62	92
171,403	BD	88	88
171,401	BD	75	88
171,404	BD	73	88
171,402	BD	65	88
171,403	ML	80	80

3.

Mahasiswa yang Mengambil SEMUA Mata Kuliah Wajib

```

SELECT m.nim, m.nama
FROM mahasiswa m
WHERE NOT EXISTS (
  SELECT 1
  FROM matakuliah mk
  WHERE mk.wajib = 1
  AND NOT EXISTS (
    SELECT 1
    FROM nilai n
    WHERE n.nim = m.nim          -- Korelasi luar
          AND n.kode_matakuliah = mk.kode      -- Korelasi dalam
  )
);

```

123 nim	AZ nama
171,401	Andi Kurniawan
171,402	Budi Santoso
171,403	Citra Dewi

Tugas Praktikum

Langkah	Jawaban/Deskripsi																
1.	Mahasiswa yang Pernah Mendapat Nilai 'A' (Menggunakan EXISTS) <pre>SELECT m.nim, m.nama FROM mahasiswa m WHERE EXISTS (SELECT 1 FROM nilai n WHERE n.nim = m.nim AND n.nilai_huruf = 'A');</pre> mahasiswa 1 X SELECT m.nim, m.nama FROM mahasiswa m Enter a SQL expression to filter results (use Ctrl+Space) <table><thead><tr><th></th><th>123 nim</th><th>A-Z nama</th><th></th></tr></thead><tbody><tr><td>1</td><td>171,401</td><td>Andi Kurniawan</td><td></td></tr><tr><td>2</td><td>171,403</td><td>Citra Dewi</td><td></td></tr></tbody></table>		123 nim	A-Z nama		1	171,401	Andi Kurniawan		2	171,403	Citra Dewi					
	123 nim	A-Z nama															
1	171,401	Andi Kurniawan															
2	171,403	Citra Dewi															
2.	Mahasiswa yang TIDAK Pernah Mendapat Nilai 'A' (Menggunakan NOT EXISTS) <pre>SELECT m.nim, m.nama FROM mahasiswa m WHERE NOT EXISTS (SELECT 1 FROM nilai n WHERE n.nim = m.nim AND n.nilai_huruf = 'A');</pre> mahasiswa 1 X SELECT m.nim, m.nama FROM mahasiswa m Enter a SQL expression to filter results (use Ctrl+Space) <table><thead><tr><th></th><th>123 nim</th><th>A-Z nama</th><th></th></tr></thead><tbody><tr><td>1</td><td>171,402</td><td>Budi Santoso</td><td></td></tr><tr><td>2</td><td>171,404</td><td>Dian Pratiwi</td><td></td></tr><tr><td>3</td><td>171,405</td><td>Eko Prasetyo</td><td></td></tr></tbody></table>		123 nim	A-Z nama		1	171,402	Budi Santoso		2	171,404	Dian Pratiwi		3	171,405	Eko Prasetyo	
	123 nim	A-Z nama															
1	171,402	Budi Santoso															
2	171,404	Dian Pratiwi															
3	171,405	Eko Prasetyo															

3.

Rata-rata Nilai Mata Kuliah > 70 (Menggunakan *Derived Table* di FROM)

```

SELECT mk.kode, mk.nama, ringkasan.rata_nilai
FROM (
    SELECT kode_matakuliah, AVG(nilai_angka) AS rata_nilai
    FROM nilai
    GROUP BY kode_matakuliah
) AS ringkasan
JOIN matakuliah mk ON mk.kode = ringkasan.kode_matakuliah
WHERE ringkasan.rata_nilai > 70;

```

matakuliah 1

SELECT mk.kode, mk.nama, ringkasan.rata_nilai

	AZ kode	AZ nama	123 rata_nilai
1	ASD	Algoritma & Struktur Data	77.75
2	BD	Basis Data	75.25
3	ML	Machine Learning	80
4	PBO	Pemrog. Berorientasi Objek	80
5	WEB	Pemrograman Web	84.5

4.

Mahasiswa dengan Nilai di Atas Rata-rata Kelasnya (Menggunakan *Correlated Sub-Query*)

```

SELECT n1.nim, n1.kode_matakuliah, n1.nilai_angka
FROM nilai n1
WHERE n1.nilai_angka > (
    SELECT AVG(n2.nilai_angka)
    FROM nilai n2
    WHERE n2.kode_matakuliah = n1.kode_matakuliah
);

```

nilai 1

SELECT n1.nim, n1.kode_matakuliah

	123 nim	AZ kode_matakuliah	123 nilai_angka
1	171,401	ASD	85
2	171,401	PBO	90
3	171,403	ASD	92
4	171,403	BD	88
5	171,403	WEB	91

5.

Mahasiswa yang Mengambil SEMUA Mata Kuliah Wajib (Menggunakan *Double NOT EXISTS*)


```
SELECT mk.nama AS nama_matakuliah
FROM matakuliah mk
WHERE NOT EXISTS (
  SELECT 1
  FROM nilai n
  WHERE n.kode_matakuliah = mk.kode
  AND n.nim = 171402
);
```

	AZ nama_matakuliah
1	Machine Learning
2	Pemrograman Web

8. Nilai Mahasiswa yang Lebih Tinggi dari Semua Nilai Kelas 'ML' (Menggunakan ALL)

```
SELECT DISTINCT m.nim, m.nama
FROM mahasiswa m
JOIN nilai n ON m.nim = n.nim
WHERE n.nilai_angka > ALL (
  SELECT nilai_angka
  FROM nilai
  WHERE kode_matakuliah = 'ML'
);
```

	123 nim	AZ nama
1	171,401	Andi Kurniawan
2	171,403	Citra Dewi